

Laporan Modul 3 — Processes, Services, Job Scheduling & Observability

Workshop Administrasi Jaringan · PENS — Teknik Informatika dan Komputer

- Kelompok: X = 2 · Fokus host: **srv01** · Akses: SSH key dari adm01
- OS: Debian 13.4 (trixie) · Tanggal: 2026-06-12

1. Ringkasan

Pada srv01 dilakukan observasi proses & resource, pengelolaan service dengan systemd, pembuatan custom service (`demo-app`), penjadwalan dengan systemd timer (serta cron sebagai pembanding), dan stress test CPU untuk observability. Semua artefak diperbarui di repo `lab-admin/`.

2. Observasi Proses (4.1)

- PID 1 = **systemd** (induk semua proses userland).
- Parent sshd = systemd; tiap sesi membuat child `sshd-session`.
- `ps` (snapshot) vs `top` (real-time) vs `htop` (interaktif).

Evidence: `evidence/process_snapshot.txt` · Catatan: `docs/process_observation_notes.md`.

3. systemd Service Management (4.2 & 4.3)

Custom service `demo-app` dibuat dari `log-demo.sh`:

```
[Service]
Type=simple
ExecStart=/usr/local/bin/log-demo.sh
Restart=on-failure
User=root
```

- `daemon-reload` → `start` → `enable`.
- Status: **enabled** (aktif saat dipanggil, lalu exit 0 — wajar untuk script singkat).
- Log via journald: `journalctl -u demo-app`.

Evidence: `evidence/service_status.txt` · Unit: `configs/demo-app.service`.

4. systemd Timer (4.4)

- `demo-app-periodic.timer` (`OnBootSec=1min`, `OnUnitActiveSec=1min`) memanggil `demo-app-periodic.service` (`Type=oneshot`).
- Status: **active + enabled**, terlihat di `systemctl list-timers` (NEXT/LAST terisi).
- Terbukti berjalan: `/var/log/demo-app.log` bertambah otomatis tiap menit (timestamp baru tiap baris).

Evidence: `evidence/timer_status.txt` · Unit: `configs/demo-app-periodic.timer`, `configs/demo-app-periodic.service`.

5. Cron sebagai Pembanding (4.5)

```
* /2 * * * * /usr/local/bin/log-demo.sh
```

Analisis cron vs systemd timer ada di `docs/scheduling_policy_v1.md` (kelebihan timer: integrasi journal, dependency, status terlihat, persistensi).

6. Resource Stress Test (4.6)

- `stress --cpu 1 --timeout 30` → 1 core 100% (%CPU 100 pada proses stress).
- Load average naik (0.34 → 0.60) selama beban.
- Analisis load average & bottleneck: `docs/process_observation_notes.md`.

Evidence: `evidence/stress_snapshot.txt`.

7. Definition of Done (Checklist)

Item	Status
demo-app.service aktif & enable	OK (enabled)
demo-app-periodic.timer jalan	OK (active, list-timers terisi)
Log terus bertambah otomatis	OK (per menit, timestamp baru)
Stress test terbukti & dianalisis	OK
Repo terupdate + evidence lengkap	OK

8. Bukti Screenshot

[1] Observasi proses (PID 1 = systemd, hierarki, konsumen memori)

```
-- [1] OBSERVASI PROSES ==
-- PID 1 --
  PID PPID COMMAND
  1    0 systemd
-- hierarki (head) --
UID    PID    PPID C STIME TTY          TIME CMD
root    2      0  0 Jun12 ?      00:00:00 [kthreadd]
root    3      2  0 Jun12 ?      00:00:00 \_ [pool_workqueue_release]
root    4      2  0 Jun12 ?      00:00:00 \_ [kworker/R-kvfree_rcu_reclaim]
root    5      2  0 Jun12 ?      00:00:00 \_ [kworker/R-rcu_gp]
root    6      2  0 Jun12 ?      00:00:00 \_ [kworker/R-sync_wq]
root    7      2  0 Jun12 ?      00:00:00 \_ [kworker/R-slab_flushwq]
root    8      2  0 Jun12 ?      00:00:00 \_ [kworker/R-netns]
-- top memori --
USER    PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root    278  0.0  0.8  51012 17944 ?        Ss   Jun12   0:00 /usr/lib/systemd/systemd-journald
root     1  0.0  0.7  24072 15272 ?        Ss   Jun12   0:03 /sbin/init
fishir  1338  0.0  0.6  22048 12352 ?        Ss   Jun12   0:00 /usr/lib/systemd/systemd --user
root    335  0.0  0.4  35268 10060 ?        Ss   Jun12   0:04 /usr/lib/systemd/systemd-udevd
fishir@srv01:~$
```

[2] Custom service demo-app (enabled + journal)

```
== [2] CUSTOM SERVICE demo-app ==
active: inactive
enabled: enabled
-- Journal demo-app --
Jun 12 02:45:04 srv01 systemd[1]: Started demo-app.service - Demo Logging Application.
Jun 12 02:45:04 srv01 systemd[1]: demo-app.service: Deactivated successfully.
Jun 12 02:45:27 srv01 systemd[1]: Started demo-app.service - Demo Logging Application.
Jun 12 02:45:27 srv01 systemd[1]: demo-app.service: Deactivated successfully.
fishir@srv01:~$
```

[3] systemd timer aktif + log tumbuh otomatis per menit

```
== [3] SYSTEMD TIMER ==
active: active
enabled: enabled
-- list-timers --
NEXT                LEFT LAST                PASSED UNIT                ACTIVATES
Sat 2026-06-13 00:47:23 WIB    407ms Sat 2026-06-13 00:46:23 WIB    59s ago demo-app-periodic.timer    demo-app-periodic.service
-- log tumbuh (timestamp per menit) --
2026-06-13T00:42:46+07:00 demo-app: hello from srv01 pid=4163
2026-06-13T00:43:45+07:00 demo-app: hello from srv01 pid=4184
2026-06-13T00:45:13+07:00 demo-app: hello from srv01 pid=4204
2026-06-13T00:46:23+07:00 demo-app: hello from srv01 pid=4221
fishir@srv01:~$ _
```

[4] Cron sebagai pembanding

```
== [4] CRON (pembanding) ==
-- crontab -l --
*/2 * * * * /usr/local/bin/log-demo.sh
-- cron service --
active
fishir@srv01:~$ _
```

[5] Stress test CPU (proses stress ~100% CPU, load average naik)

```
== [5] STRESS TEST CPU ==
-- load SEBELUM --
00:48:43 up 3:45, 1 user, load average: 0.16, 0.04, 0.01
[1] 4264
stress: info: [4264] dispatching hogs: 1 cpu, 0 io, 0 vm, 0 hdd
-- top SAAT stress --
top - 00:48:50 up 3:45, 1 user, load average: 0.23, 0.06, 0.02
Tasks: 112 total, 2 running, 110 sleeping, 0 stopped, 0 zombie
%Cpu(s): 40.0 us, 5.0 sy, 0.0 ni, 55.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1973.7 total, 1630.7 free, 261.2 used, 217.8 buff/cache
MiB Swap: 1101.0 total, 1101.0 free, 0.0 used, 1712.5 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
  4266 fishir    20   0   3524    512   396 R   90.9    0.0   0:06.21 stress
  4269 fishir    20   0  10224   5220  3208 R    9.1    0.3   0:00.01 top
-- load SAAT --
00:48:50 up 3:45, 1 user, load average: 0.23, 0.06, 0.02
fishir@srv01:~$
```

9. Penyimpangan dari Modul

- OS Debian 13 (trixie), bukan Debian 12 — mengikuti ISO yang tersedia.
- User admin `fishir` menggantikan `student`.